Proof-Carrying Detection of Reticulate Evolution: a Tree-Level Null, the Measured Cost of the Alignment Step, and the Gromov Hinge of Buneman's Converse

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Abstract

Tests for recombination are normally calibrated against a permutation ensemble. We develop one calibrated against a *theorem*, report the two ways our own first version of it was wrong, and give the measurement that we believe is new.

A strict molecular clock on a clonal genealogy makes expected p-distances ultrametric, and an ultrametric admits no unfilled Vietoris–Rips cycle at any scale; three sequences violating the strong triangle inequality are therefore a *certificate*, not a surprise relative to shuffles. We formalise the inference rule in Lean 4 and emit certificates that embed the aligned rows, so the kernel recomputes every count by `rfl` rather than trusting a search.

The statistic this licenses is nonetheless the wrong one, and a control we had not run says so. On a clonal tree with a **relaxed clock and no reticulation whatever**, divergence-matched so that only the clock differs, the ultrametric margin exceeds every single-recombinant margin in the positive control; at 32 leaves $\min(\text{no reticulation}) = 99/4000 > 79/4000 = \max(\text{one recombinant})$ at every seed. The four-point condition encodes *tree* without *clock*, we prove in Lean that every symmetric ultrametric satisfies it, and the quartet margin on the same data does not move. So the quartet statistic must carry the method, and the licence for reading the gap between the two conditions is a theorem rather than a convention.

Running this on real data required writing the alignment step, which puts an aligner on the trust surface that no downstream proof covers. We measure it instead of caveating it: on simulated populations with no reticulation and a known true alignment, reference-anchored alignment manufactures 0 of 9100 four-point violations at a 0.4% gap load and 1600 of 9100 at 8.2%. We find no published benchmark that varies alignment quality as an independent variable, so we believe this is the first direct measurement of a mechanism the field already asserts. Read as an admissibility threshold it refuses 7 of 10 real viral populations drawn from a public corpus; the 3 that pass agree with a literature prior fixed in advance, and the two certificates clearing their matched floor are accepted by the kernel with both corrupted twins rejected.

Two corrections to our own apparatus are reported as results. The band-edge test by which we had certified the aligner is not a soundness certificate, and it had really failed: a pair scoring 16555 against an unbanded optimum of 17275 with the band edge never touched. Certifying every alignment against the unbanded optimum dropped the figure above from 4876 to 1600 — two thirds of our headline was an uncertified step inside the measuring apparatus. Separately, the statistic we report is a maximum over subsets where the established treelikeness tests are averages; run seed-paired against the hard null, the maximum separates 16.57× against 5.07× for mean δ_q and 1.22× for the proportion of violating quadruples.

Finally we prove the hinge of Buneman's converse, which this programme had only cited: a symmetric dissimilarity satisfies the four-point condition if and only if its Gromov product satisfies the max inequality at every root. We find no formalisation of tree metrics, splits or Buneman's theorem in Lean, Isabelle or Coq, so we believe this is first of its kind. All Lean declarations compile with zero `sorry` and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}.

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1 Introduction

A test for recombination has to answer the question: compared with what? The standard answer is an ensemble — shuffle the columns of the alignment, recompute the statistic, and read off a p-value. It is a serviceable answer with two costs. The first is a resolution floor: with N shuffles no p-value below $1/(N+1)$ is attainable, so the apparatus has an opinion about what counts as significant before the data arrives. The second is sharper and less often stated. The manual of the field's dominant tool observes that misalignment produces false recombination signals which *"will be detected as recombination events with significant p-values"* while *"these false positive signals will likely be undetectable in the shuffled alignments"* [10]. Column shuffling destroys the spatial clustering of misalignment along with the signal, so an entire class of artefact survives the real analysis and vanishes from the null. A permutation null cannot see it.

This paper develops the alternative: a null that is a theorem, so that a violation is a witness rather than a tail probability.

Modelling assumption 1.1 (What the clock-level null assumes)**.** Write $d(x, y)$ for the p-distance between two aligned sequences — differing columns over columns comparable in both. Expected p-distances satisfy

$$d(x, z) \;\le\; \max\big(d(x, y),\, d(y, z)\big) \qquad \text{for all } x, y, z \tag{1}$$

exactly when descent is **clonal** *and* the molecular clock is **strict**. Two assumptions, one inequality.

If (1) holds then $d(\cdot,\cdot) \le \epsilon$ is transitive, so every connected component of the Vietoris–Rips complex at scale ϵ is a clique and no unfilled cycle exists at any scale. Three sequences with $\max(d(x,y), d(y,z)) < d(x,z)$ therefore contradict a universally quantified statement. Nothing is calibrated.

Section 2 states the two nulls and proves they nest. Section 3 refutes the clock-level statistic with a control that contains no reticulation. Section 4 deals with missing data. Section 5 is the paper's main empirical contribution: the alignment step is on the trust surface, and this is how large it is. Section 6 reports the proxy inside our own apparatus that failed and what it had inflated. Section 7 prices the maximum against the average. Section 8 runs the test on real viral populations. Section 9 proves the hinge of Buneman's converse. Section 10 says what would make all of it wrong.

What is and is not claimed. We claim: the nesting theorem and the Gromov equivalence, both machine-checked; the refutation of the clock-level statistic on simulated data; the measurement of alignment-manufactured violations; and three real-data verdicts that agree with priors fixed in advance. We do not claim a validated method. Three admissible populations is not a validation, no head-to-head against the established tests has been run, and Buneman's converse remains cited for its second half. This paper has not been peer reviewed.

2 Two nulls, and the fact that they nest

Definition 2.1 (Four-point condition)**.** A symmetric d satisfies the *four-point condition* when for all a, b, c, e, not necessarily distinct,

$$d(a,b) + d(c,e) \;\le\; \max\big(d(a,c) + d(b,e),\; d(a,e) + d(b,c)\big). \tag{2}$$

A quadruple admits three pairings; (2) says no pairing's distance sum strictly exceeds both others. This is a statement about tree structure with no constraint on rates: branch lengths are free.

Theorem 2.2 (Tree-Metric Theorem; Buneman. *Cited, not proved here*)**.** *Let δ be a nonnegative dissimilarity map on a finite set — zero on the diagonal, symmetric, non-negative. Then δ is a tree metric if and only if it satisfies* (2) *[1, 2].*

The hypotheses are weaker than "metric", and the distinction matters enough to state: setting $y = z$ in (2) yields the triangle inequality, so the triangle inequality is a *consequence* of the condition and requiring it as a hypothesis would state something strictly weaker than the theorem. An earlier draft of our own documentation made that error.

Theorem 2.3 (The nulls nest)**.** *Every symmetric ultrametric satisfies the four-point condition.*

Proof. Fix a, b, c, e. Four applications of (1), with symmetry to order each application, give $d(a,b) \le \max(d(a,c), d(b,c))$, $d(a,b) \le \max(d(a,e), d(b,e))$, $d(c,e) \le \max(d(a,c), d(a,e))$ and $d(c,e) \le \max(d(b,c), d(b,e))$. Each is a disjunction, giving 2^4 cases; splitting further on the order of the four bounding quantities, every branch bounds $d(a,b) + d(c,e)$ by one of the two right-hand sums of (2) by a linear arithmetic step. The case analysis is mechanical and is discharged by the kernel (Listing 1). □

Corollary 2.4 (What a quartet violation buys)**.** *If d is symmetric and fails* (2) *then d is not ultrametric either.*

Corollary 2.4 is the licence for the reading that organises Section 3: large triple margins together with floor-level quartet margins indicate rate variation on a tree, because the nesting leaves that the only region available.

Table 1: Thirty-two leaves, 4000 nt, three seeds, exhaustive over all 35 960 quadruples per dataset. Margins are exact rationals with denominator 4000; numerators given seed by seed.

simulated population	reticulation?	clock	triple margin	quartet margin
clonal	no	strict	$24, 30, 29$	$10, 9, 10$
clonal	**no**	**relaxed**	$\mathbf{155}, \mathbf{157}, \mathbf{99}$	$11, 18, 8$
one recombinant	yes	strict	$79, 77, 75$	$79, 70, 69$
two reciprocal mosaics	yes	strict	$79, 77, 75$	$\mathbf{158}, \mathbf{154}, \mathbf{126}$

Listing 1: `formal/ANSE/GeneticaReticulataTree.lean`. The `rcases` chain is the 2^4 case split, with four further splits on the order of the bounding quantities.

```
def IsFourPoint (d : α → α → ℚ) : Prop :=
  ∀ a b c e, d a b + d c e ≤ max (d a c + d b e) (d a e + d b c)

theorem fourPoint_of_ultrametric {d : α → α → ℚ}
    (hsym : ∀ x y, d x y = d y x) (hu : IsUltrametric d) : IsFourPoint d
```

The module carries 18 declarations through an axiom gate that compiles each and runs `#print axioms`: zero `sorry`, and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}. Checking compilation alone is insufficient — both `sorry` and a locally declared axiom exit zero — which is why the gate inspects the axiom list.

3 The clock-level statistic is refuted by a control with no reticulation

Condition (1) needs three sequences rather than four, so it is cheaper, and our first version reported it. It separated a simulated clonal population from a simulated recombinant cleanly. The separation was real and the inference from it was wrong, for the reason Modelling assumption 1.1 warns of: a violation tells you one of two assumptions failed, not which.

The decisive control is a **divergence-matched relaxed clock**: clonal descent on a balanced tree, no reticulation whatever, with each lineage's substitution rate multiplied by e^{su} for u uniform on $[-1, 1]$. The matching is not optional. Since $\mathbb{E}[e^{su}] = \sinh(s)/s > 1$, an unnormalised relaxed clock is also a *faster* clock, and a violation could then be attributed to saturation rather than to rate variation; dividing by that mean leaves expected divergence equal to the strict-clock case while realised rates vary widely. In our first run the normalisation changed the answer, taking the relaxed-clock quartet violation count from 2/5 seeds to 0/5.

Read rows two and three of Table 1 together. The smallest triple margin produced by a population with *no reticulation in it at all* is 99/4000; the largest produced by a population containing a real recombinant is 79/4000. The ordering is inverted at every seed. On this data a larger triple margin is weak evidence *against* reticulation. The quartet column does not move: the relaxed clock stays at the strict clock's level while the reciprocal-mosaic positive control separates from it by a factor of 12.6 at its weakest seed, with no overlap. At sixteen leaves the same experiment gives a relaxed-clock triple margin reaching 106/4000 — above every single-recombinant margin — exceeding twice the clonal floor at 3 of 5 seeds, while the quartet statistic does so at 0 of 5.

Adding taxa makes the triple statistic worse, as the mechanism predicts: it is a maximum over triples, so more leaves give more chances at an extreme pair of rates.

Remark 3.1. This confound is not peculiar to our statistic. Frankel and Ané report that rate variation across lineages produces a marked increase in the false discovery of reticulation for three summary tests of introgression, with the D_3 statistic reaching about 80% type-I error, and

read this as D_3 responding more strongly to a departure from the clock than to reticulation itself [8]. A null that assumes a clock is vulnerable wherever it is used. The established test whose null assumes no clock, Bruen, Philippe and Bryant's Φ_w, reports 5.7% false positives on a non-recombinant control where NSS reports 37.5% [6].

Recommendation. Report the quartet statistic as primary. Keep the triple statistic — it is cheaper, and a triple violation with *no* quartet violation is itself informative, being the signature of rate variation on a tree — but a triple certificate alone is not evidence of reticulation.

4 Missing data: complete deletion within the subset

Pairwise deletion computes each pair's distance over the columns comparable in that pair, giving different pairs different denominators, and that alone can break (1) on a genealogy that is exactly a clock tree. The fix is *complete deletion within the subset*: count only columns at which every member of the triple or quadruple carries a definite base. All distances in a certificate then share one denominator, so the obligations become comparisons between integers with no cross-multiplication — a smaller trust surface and the repair at once. The cost is power: at a 25% gap load roughly 1000 to 1700 of 4000 columns survive within a triple.

Measured on gap-blocked but strictly clonal, strictly clock-like populations: pairwise deletion calls 33 to 52 of the 560 triples violations that complete deletion within the triple denies, at every one of five seeds; at thirty-two leaves, 286 to 366 of 4960. With no missing data the two policies agree on all 560, as they must.

Remark 4.1 (Margins under the two policies are not comparable). Complete deletion shrinks the denominator, so one differing site is worth more and its margins come out *larger* although the policy is strictly more conservative. A criterion comparing the two sets of margins was written into our first run and withdrawn. Compare per-subset *verdicts*, not magnitudes.

5 The alignment step is on the trust surface

A certificate is machine-checked, which invites the claim that the conclusion does not depend on the pipeline. The precise statement is narrower. The kernel is handed the sequences and recomputes every count from them, so a false inequality, a wrong count, a transposed pair or a miscomputed margin is excluded. What remains trusted is that the embedded rows are the data they are claimed to be — and on real data those rows come out of an **aligner**. If the alignment places non-homologous residues in the same column, the counts are computed faithfully from the wrong data and the kernel accepts them. Proof moves the trust boundary; it does not close it.

That cannot be proved away, so we measure it. We simulate clonal, relaxed-clock descent with deletion events in a fixed coordinate frame, so the frame *is* the true alignment and **there is no reticulation in the population at all**; strip the gaps; hand the ragged sequences to the aligner; realign; and compare per-quadruple *verdicts*, not margins, since the two alignments have different widths. Any quadruple the inferred alignment calls a violation and the true alignment does not was manufactured by the aligner.

The aligner is reference-anchored: each sequence is aligned to a centroid reference by affine-gap global dynamic programming and projected onto the reference's coordinates. This is the construction viral genomic surveillance uses, and the final step of MUSCLE5's `Super5` mode. Its one structural advantage is that without a guide tree there is no error-propagation failure mode; its costs are that insertions relative to the reference are dropped and that every pair of non-reference sequences is related through the reference.

At the heaviest rung the aligner invents 1600 of the 9100 quadruple verdicts on a genealogy that is exactly a tree. **Past a threshold, alignment error is not a correction to this test;**

Table 2: The admissibility ladder. Sixteen taxa, 4000 nt, five seeds, 1820 quadruples per seed (9100 per rung). Every alignment certified optimal (Section 6).

mean deletion length (nt)	inferred gap fraction	worst column recovery	quadruples manufactured	seeds with any
30	0.00000	1.00000	0	0/5
30	0.00398	0.99997	0	0/5
30	0.00549	0.99880	251	2/5
30	0.00870	0.99712	108	1/5
30	0.01507	0.99906	110	3/5
30	0.02012	0.99831	149	3/5
120	0.00000	1.00000	0	0/5
120	0.01322	0.99901	57	2/5
120	0.01627	0.99901	54	2/5
120	0.03731	0.99467	324	3/5
120	0.05254	0.97501	253	3/5
120	0.08233	0.96032	**1600**	5/5

it is the result. The curve is not monotone, which is what a maximum over 1820 quadruples at five seeds looks like, and is one reason to gate on *whether* anything was manufactured rather than on how much.

So the admissibility threshold is read off the ladder rather than chosen: a gap fraction of 0.00398 and 8.06×10^{-4} dropped insertions per aligned cell, the heaviest rung at which nothing was manufactured at any seed. Both are *rates*; an absolute insertion count is not comparable between a 3 kb and a 12 kb genome.

Why we believe this measurement is new. That misalignment manufactures recombination signal is not a novel claim and not a contested one. It is stated outright in the RDP5 manual — *"misalignment of sequences is a major cause of false recombination signals"* and *"all recombination signal detection methods are extremely sensitive to misalignment and will usually identify misaligned regions of sequence as having a recombinant origin"* — and misalignment artefact is a named output category in that program [10]. What we did not find is the quantification. Across the recombination benchmarking literature we examined, every study varies divergence, rate heterogeneity, topology, selection or sequence count while holding the alignment fixed; the most recent full benchmark we reached does not discuss gaps at all [11]. A measurement of the form "alignment error alone manufactures N of M apparent signals on data containing no reticulation" appears therefore to be the first direct measurement of a mechanism the field already accepts. That is confirmatory-but-novel, and we state it that way rather than as a discovery. The nearest quantified precedent we found is in an adjacent task: aligner choice driving a reported 48–82% of false positives in *Drosophila* positive-selection scans.

An honest limitation of the simulation. The deletion rate and mean deletion length are *chosen*, not calibrated: we found no published viral indel length distribution and no clean per-site viral indel rate. The ladder therefore varies a load; it does not match one to biology.

6 Certifying the band: a proxy inside our own apparatus

The dynamic programme is banded, because 12 kb against 12 kb is 1.4×10^8 cells and the optimal path of two homologues does not wander far from the diagonal. A band is a cap on how much indel a pair may be assigned, so it is a place where an aligner can silently return a sub-optimal path. Our first implementation detected when the optimal traceback *touched* a band edge, widened the band until it did not, and documented this as a guarantee that no constrained path was ever returned without saying so.

That documentation was wrong. Gibrat measures both failure directions on real data: 2.2% of alignments are optimal yet reach the band edge, and 0.5% are sub-optimal without ever reaching it [4]. Not touching the edge is evidence of optimality, not proof of it.

The repair needs no theorem. Computing the unbanded optimum by full Gotoh dynamic programming in $O(nm)$ time and $O(m)$ memory — no band, no heuristic, no traceback — and refusing any banded path that does not attain it gives a certificate for one extra pass per pair, which at fifteen pairs per population is affordable. We also note, and do not use, a cheaper sufficient test: Gibrat gives an upper bound on the score of any path leaving the band, so attaining that bound certifies optimality in constant time. His derivation is for a band about the main diagonal while ours is about the proportional diagonal, so the bound does not transfer unexamined, and recomputation needs no derivation to be trusted.

The proxy had really failed. The first certified run exited non-zero immediately, on a pair scoring 16555 against an unbanded optimum of 17275 — a deficit of 720 points, with the traceback never touching an edge, so edge-driven widening never fired. The widening loop is now driven by the *score*: widen on either signal, and return only a path that attains the optimum, or an error.

What it had inflated. With every alignment proved optimal, Table 2's heaviest rung manufactures 1600 rather than 4876, and worst column recovery is 0.960 rather than 0.868. Roughly two thirds of what we had reported as reference-anchored alignment error was our own unchecked heuristic. The qualitative conclusion is unchanged and the thresholds do not move, so no verdict in Section 8 changes; but the magnitude was overstated threefold, in the direction that makes a finding look more impressive. We record this because it generalises: **an uncertified step inside a measuring apparatus inflates the thing being measured**, and it will tend to inflate it favourably, which is the direction least likely to prompt a second look.

Remark 6.1 (A textbook algorithm that is wrong in print). While checking the above we found that Gotoh's *published* initialisation is incorrect — it omits one of two gap-opens and can return a score that is not the cost of any alignment — and that the error is reproduced in standard textbooks and in five of ten inspected implementations [5]. Ours turns out to be the corrected form, but nothing in our codebase established that until the dynamic programme was tested against exhaustive enumeration of every alignment of short sequences rather than against another dynamic programme.

Remark 6.2 (A tie-break that is a free parameter). The control that the aligner is the identity on gapless input failed at one seed of five on our first implementation, and not because of a bug. With an affine gap of -10 to open and -1 per residue, three consecutive mismatches score $3 \times (-4) + 5 = -7$, and deleting one reference residue, aligning the rest, then inserting one query residue scores $-11 + 15 - 11 = -7$. The identity was one of two optimal alignments and the implementation returned the other. The repair was to *state* the preference. An aligner has free parameters that no amount of downstream proof constrains.

7 A maximum, where the established tests are averages

Our certificate is a maximum over quadruples: the single most-violating quadruple, exhibited as a witness the kernel can check. That is what makes a certificate possible, and it is also an extreme-value statistic whose null level grows with the number of subsets examined. Our own grid shows it — the quartet floor moves from 5–7 at sixteen taxa to 9–10 at thirty-two — so margins are not comparable across populations of different size.

The treelikeness literature took the other branch: its statistics are averages over quadruples, among them $\delta_q = (m_1 - m_2)/(m_1 - m_3)$ for sorted pairing sums $m_1 \geq m_2 \geq m_3$ [7]. We therefore ran the comparison.

Table 3: Seed-paired against the hard null (clonal, relaxed clock, no reticulation). Separation is the weakest seed's ratio of the reciprocal-mosaic control to that null.

statistic	what it is	beats the null	worst-seed separation
quartet max margin	exact, with a witness	5/5	**16.57**×
mean δ_q	the averaged statistic	5/5	5.07×
proportion violating	share of quadruples	5/5	1.22×

All three rank both reticulate populations above the relaxed-clock null at every seed, so none of them inherits the confound that destroyed the triple statistic. At fixed taxon count the maximum separates more than three times as widely as the mean, and the proportion-based variant barely separates at all. We note that this reproduces, independently, a published finding that the proportion-of-resolved-quadruples variant fails as a test while the averaged variants hold a correct false-positive rate.

What the comparison does *not* settle is the maximum's actual weakness, which is incomparability across taxon counts rather than poor separation within one; our grid is all at one size. Both statistics are therefore reported, and only the maximum is exact and kernel-checkable, because a mean has no witness to exhibit. That is the trade, and it is now priced rather than assumed.

8 Real populations

Five viral taxa were drawn from a public corpus of 58 046 viral genomes with taxonomy [12] under filters fixed in advance: within 2% of the taxon's median length, at most 0.1% ambiguous residues, exact duplicates removed, sixteen sequences per population, selected either in accession order or by greedy 6-mer diversity. Each taxon carries a *literature* expectation about whether its genealogy is tree-like, written before any margin was computed, and the test is free to contradict it. Ten populations result.

Seven fail the admissibility gate of Section 5 and are reported blocked with their counts rather than interpreted. This is expensive on purpose: one discarded population has a margin of 30/1784 and another 13/5082. Note also which selection strategy survives — diversity maximisation is exactly what breaks reference-anchored alignment. Each of the three that pass is aligned twice, on the two most central sequences, and each margin is compared against a floor from a simulated clonal relaxed-clock tree at *matched* alignment width, *matched* mean p-distance (tuned by bisection) and *matched* gap load, because a margin alone is not a statistic and a floor from the wrong divergence measures the floor.

Three populations, three agreements with the prior. The JC polyomavirus row is the one that could most easily have embarrassed the method and did not: its margin is a sixth of the floor a tree produces at matched width and divergence. Both populations clearing their floor

Table 4: The three admissible populations. References agree on the verdict in all three.

population	expectation	margin	matched floor	verdict
Hepatitis B, accession order	reticulate	9/3182	2/3182	above 2×
Coxsackievirus A6, diversity	reticulate	329/7312	268/7390	above 1× only
JC polyomavirus, accession order	tree-like	4/5108	24/5121	**below**

carry a Lean file embedding the four aligned rows, so the kernel recomputes the six pair counts and the common column count itself: accepted at 3182 columns in 52.7 s and at 7434 columns in 117.6 s, with both corrupted twins rejected.

Two cautions belong beside Table 4 rather than in a footnote. Three is not a validation. And the Coxsackievirus row is weak and at high divergence (mean $p = 0.11$): it clears 1× and not 2×, and the noise floor itself grows with divergence, so it should be read as not inconsistent with the prior rather than as a detection. A further caution concerns the prior itself: one of our two negative-control families, the papillomaviruses, has documented within-type recombination [9], so that prior is the weakest in the set and we mark it as such. A clonal bacterial lineage would be the stronger negative control.

9 The Gromov hinge of Buneman's converse

Theorem 2.2 is cited above and not proved. The constructive proof in the literature does not attack the tree directly: it factors through the *Gromov product* at a chosen root, reducing the four-point condition on d to an ultrametric-style condition on a derived function, and only then builds an equidistant tree by a cherry-picking recursion. The reduction is the hinge, and unlike the tree construction it is pure algebra.

Definition 9.1. At a root r, write $g_r(x, y) = \frac{1}{2}\big(d(x, y) - d(r, x) - d(r, y)\big)$. This is a dissimilarity, not a metric: it takes negative values. Say f satisfies the *strong triangle inequality* when $f(x, z) \le \max(f(x, y), f(y, z))$ for all x, y, z — inequality (1) with no positivity required.

Theorem 9.2 (The hinge; proved, not cited)**.** *A symmetric d satisfies the four-point condition if and only if g_r satisfies the strong triangle inequality for every root r.*

Proof. Expanding $g_r(x, z) \le g_r(x, y)$ gives $d(x, z) + d(r, y) \le d(x, y) + d(r, z)$, and $g_r(x, z) \le g_r(y, z)$ gives $d(x, z) + d(r, y) \le d(y, z) + d(r, x)$. So the right-hand side of the theorem says exactly

$$d(x, z) + d(r, y) \;\le\; \max\big(d(x, y) + d(r, z),\; d(y, z) + d(r, x)\big)$$

for all r, x, y, z, which is (2) instantiated at $(a, b, c, e) = (x, z, r, y)$ together with the symmetry of d. Reading the implication backwards requires the root c with the remaining three in the order (a, e, b). □

Listing 2: The Lean statement. `#print axioms` gives `[propext, Classical.choice, Quot.sound]`.

```
theorem fourPoint_iff_gromov_strongTriangle {d : α → α → ℚ}
    (hsym : ∀ x y, d x y = d y x) :
    IsFourPoint d ↔ ∀ r, StrongTriangle (gromov d r)
```

Two details are worth recording. The condition is named `StrongTriangle` and not `IsUltrametric`, because the Gromov product takes negative values and calling it an ultrametric would import a positivity hypothesis it does not satisfy; the literature's phrase "the Gromov product is an ultrametric" means only the max inequality. And the kernel earned its

place: our first attempt instantiated the reverse direction at the wrong permutation, `linarith` failed on two goals, and the axiom gate reported `sorryAx` rather than letting a plausible-looking proof through.

What remains cited is the tree construction — the equidistant cherry-picking recursion and the re-rooting that returns a tree metric. The chain this paper can claim is therefore: certificate ⇒ not four-point ⇒ (proved) some root's Gromov product fails the strong triangle inequality ⇒ (cited) not realised by a weighted tree. We searched Mathlib, the Isabelle Archive of Formal Proofs and the Coq ecosystem and found ultrametric spaces but no tree metrics, no splits and no four-point condition, so completing the construction appears to be first of its kind. That search was across indexes rather than exhaustive.

10 Limitations, and what would make this wrong

- **Buneman's converse is half-cited.** Theorem 9.2 closes the first step; the construction is not formalised. So "satisfies (2) everywhere" is not yet formally "is a tree" in this development.
- **A certificate does not identify an event.** Satisfying (2) does not certify the *absence* of reticulation, and a violation does not pin down which tree the data departs from; there exist networks whose distances are representable by a tree other than their own underlying tree [3]. The honest report is *that* the data are not tree-like and *where*, not an event history.
- **The margin's null distribution is not derived.** Finite sequences are never exactly ultrametric, so certificates exist under the null with small margins; we calibrate by matched mechanistic simulation rather than by a distribution. We found no analytic sampling distribution of the four-point margin in the literature either, only high-probability sequence-length bounds and parametric bootstrap, so this is the field's position and not only ours.
- **Three admissible real populations.** Too few to call the method validated, and the aligner that made them admissible is reference-anchored.
- **No head-to-head.** The comparison against Φ_w and the quartet-averaged treelikeness tests on the same alignments has not been run. Until it has, this method's standing relative to the incumbents is unestablished.
- **The statistic is an extreme value.** Section 7 prices this at fixed taxon count only.

What would make the programme wrong, rather than merely incomplete: if violations of (2) at the scale we report turned out to be dominated by some departure from tree-likeness other than reticulation — population structure, recurrent mutation, or selection acting locally — then the certificates would remain valid and the biological reading of them would not. We have controlled for rate variation and for missing data, and measured the alignment contribution. We have not controlled for the rest.

11 Reproducibility

Every number in this paper is produced by a named command in the repository `SocrateAI-Scientific-Genetic` at release `v1.8.0`, and the raw output is deposited with it.

command	what it produces
`reticulation experiment`	Table 1, Section 4, Table 3
`reticulation align-error`	Table 2 and the admissibility threshold it measures
`reticulation real-data`	Table 4 and the kernel verdicts
`reticulation saturation`	the length-scaling diagnostic behind the divergence caution
`scripts/reticulata_hf_populations.py`	the ten populations, with a SHA-256 of the source file
`lake build ANSE.GeneticaReticulataTree`	the theorems, with `#print axioms` on each

All are deterministic given their seeds. The population extraction is deterministic given the source corpus, whose digest, filters and every selected accession are recorded in `provenance.json`. Verification at the release: 88 library and 2 integration tests in Rust; Lean 18/18 and 14/14 declarations with zero `sorry` and no unwhitelisted axioms; 33 Python tests; an anti-stub audit over 227 files.

Status of this manuscript. Not peer reviewed. The repository previously described work as "5-round Opus peer reviewed" on the basis of a function that made no model call; that description was withdrawn, and no equivalent claim is made here. Reported results are the output of the commands above, including the three places where that output contradicted what we had written.

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